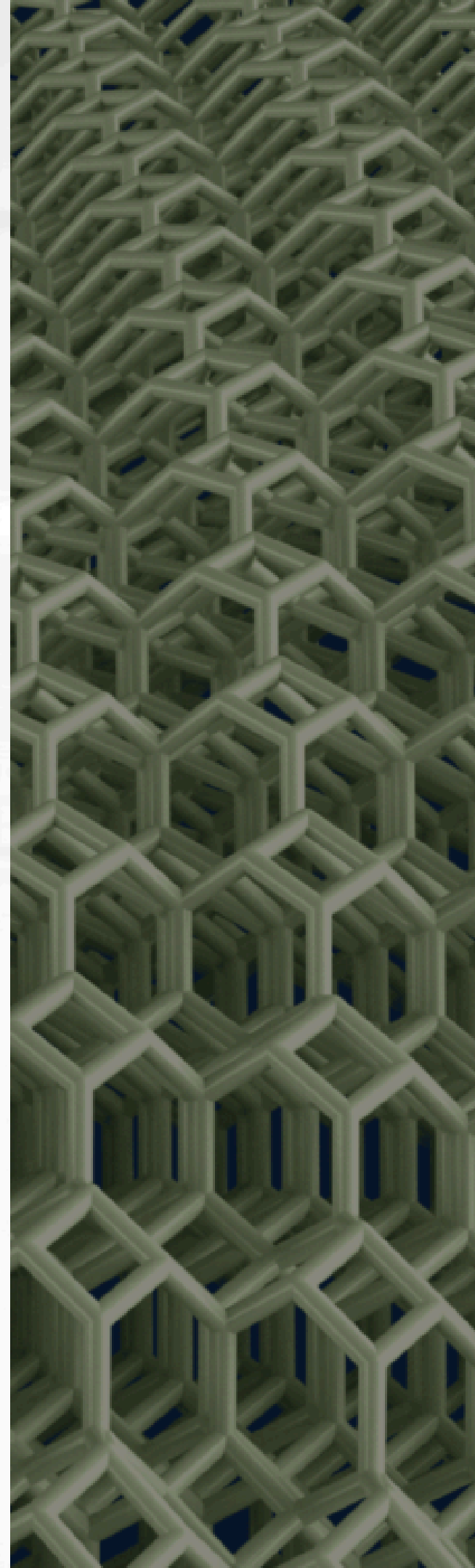
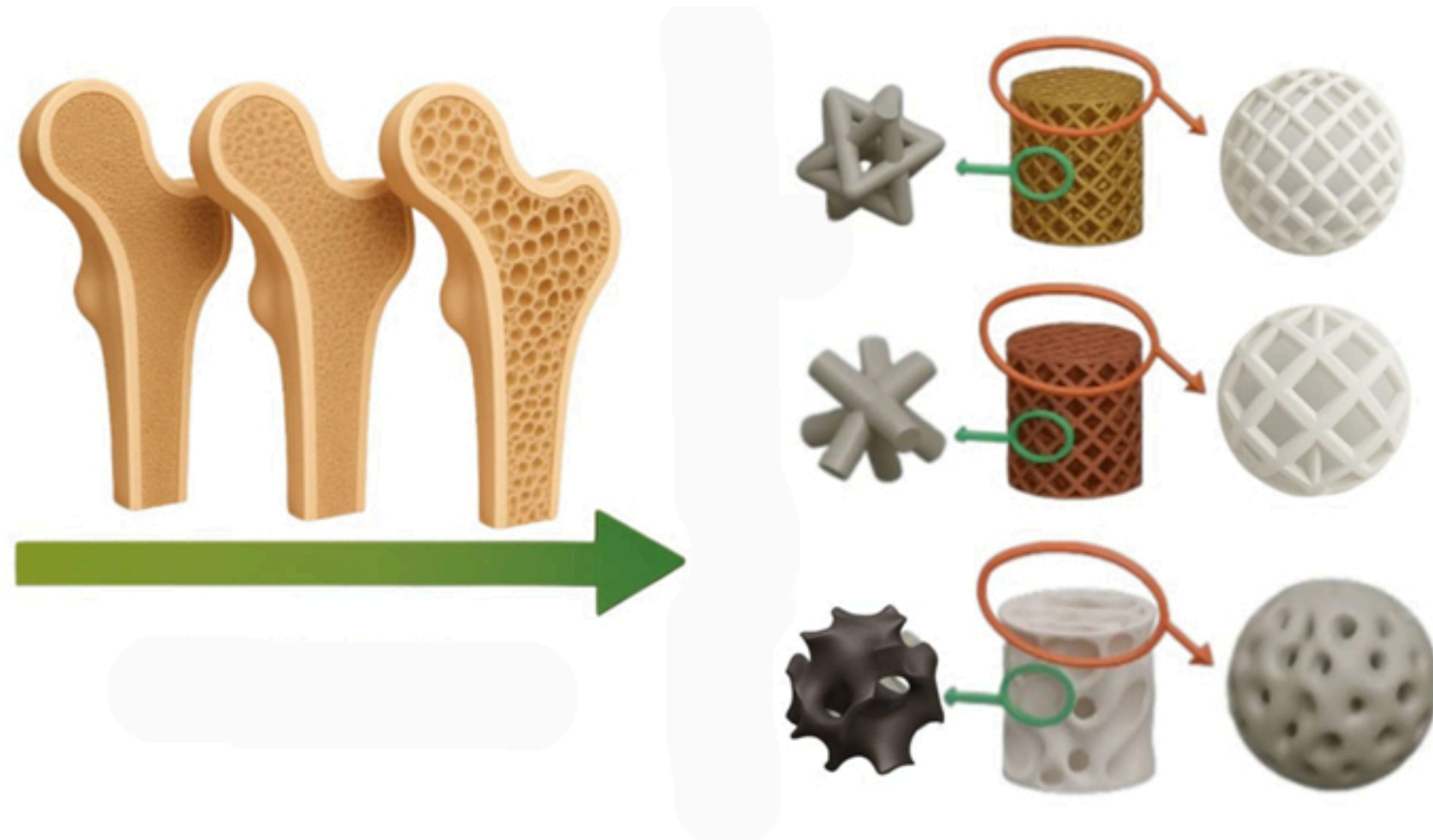


Optimization of lattice

Thomas Kuntze
Maud Faivre d'Arcier
Mathélio Favre-Réguillon
Manuel Pascual



Introduction



Design Problem

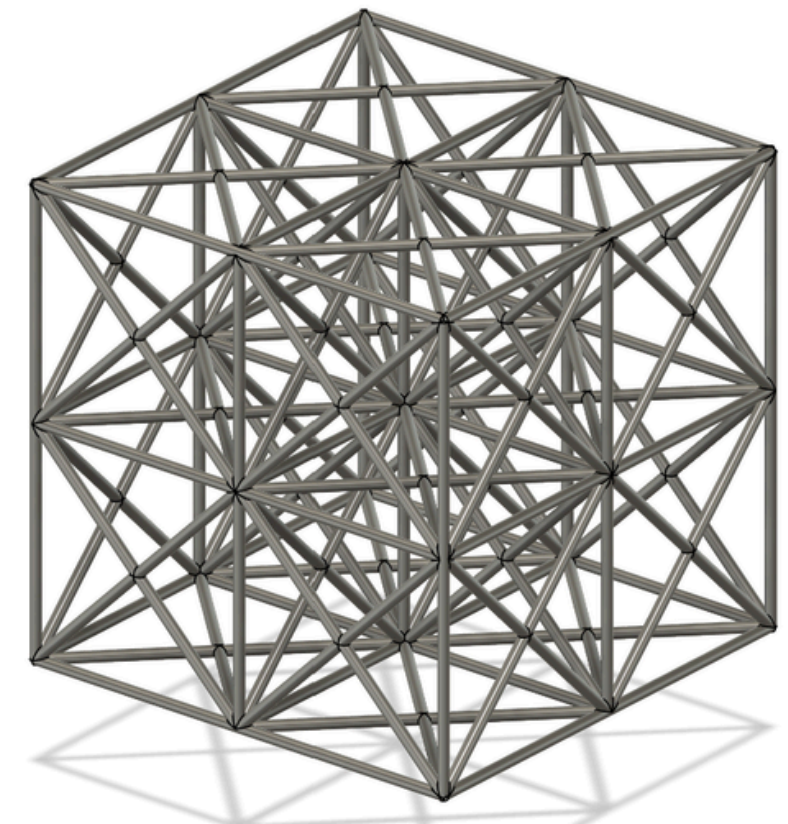
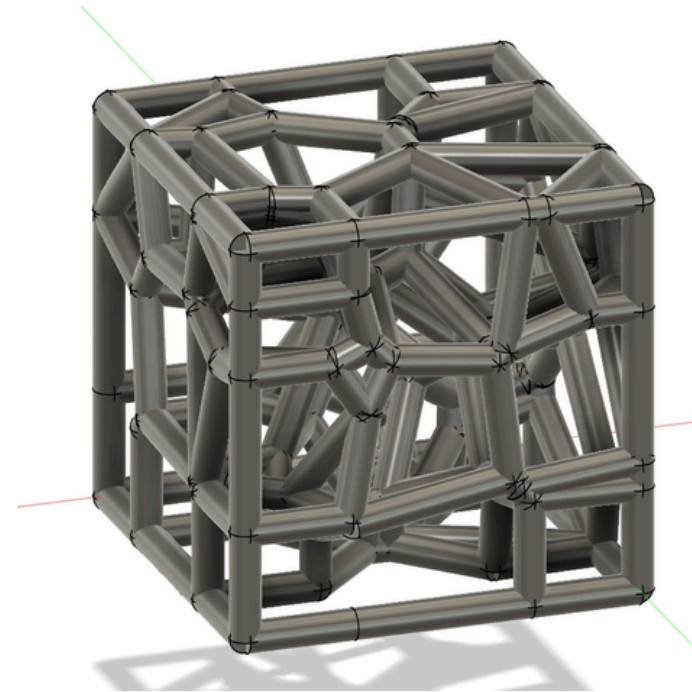
PROBLEM FORMULATION

$$\min_{\mathbf{x}} \mathbf{F}(\mathbf{x}) = [m(\mathbf{x}), \sigma_{VM}^{\max}(\mathbf{x})]$$

STUDY CASES

Load magnitudes

- 25 kg
- 50 kg
- 75 kg



Data Generation



A Python-based workflow for parametric lattice generation and mechanical simulation

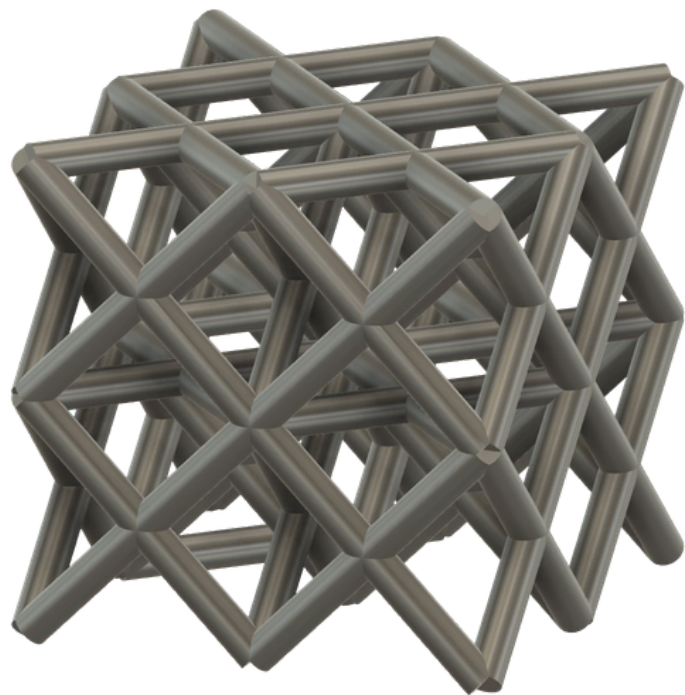
1 GEOMETRY GENERATION

Parametric **CAD generation** of lattice structures:

- Definition of unit cell size and number of repetitions
- Control of strut radius and optional reinforcements



.step



Data Generation



A Python-based workflow for parametric lattice generation and mechanical simulation

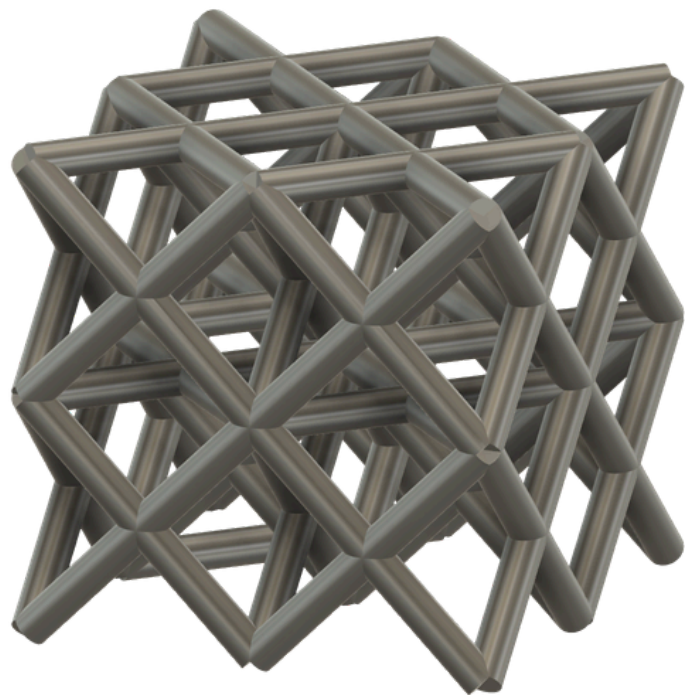
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.step



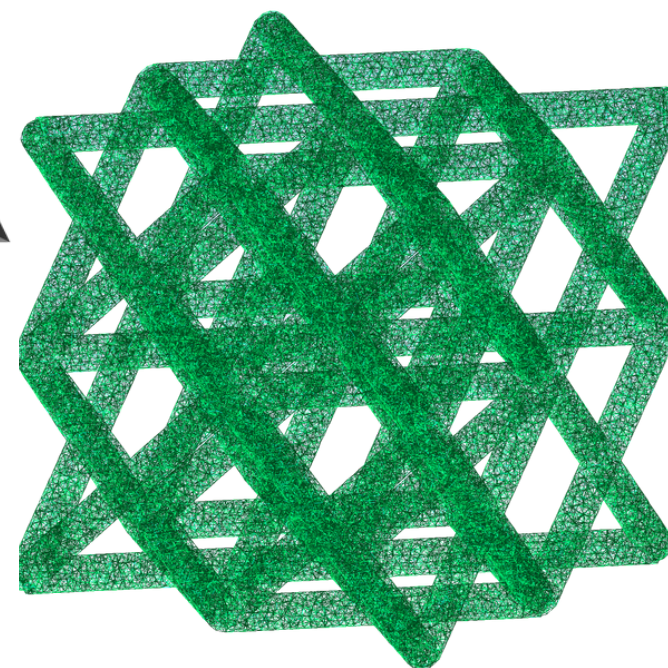
2 MESH GENERATION

Automatic meshing for finite element analysis:

- Tetrahedral finite elements
- Use of **Gmsh** for robustness : size parameter accuracy vs computational cost



.msh



Data Generation



A Python-based workflow for parametric lattice generation and mechanical simulation

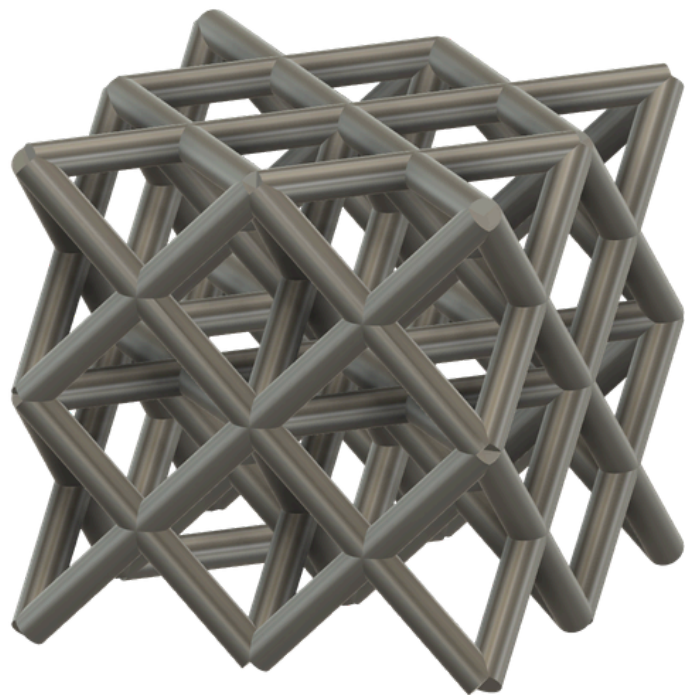
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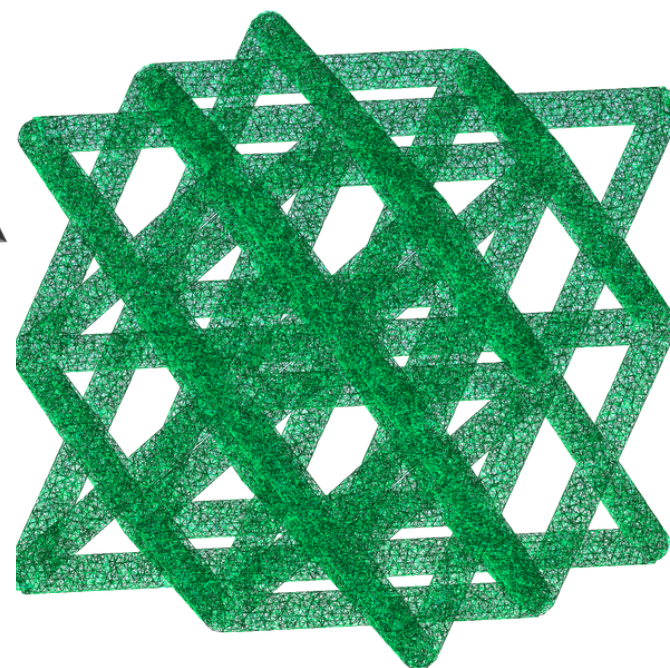
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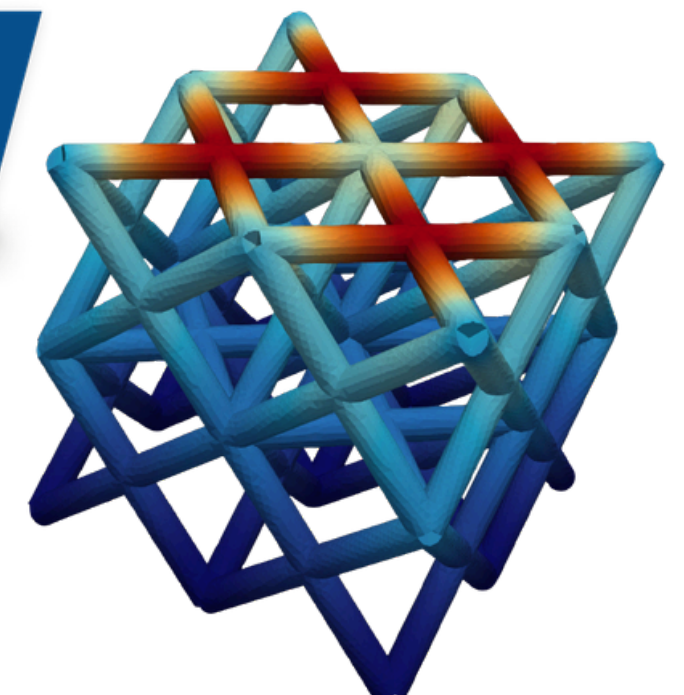
3 SIMULATION MODEL

Simulation-ready model :

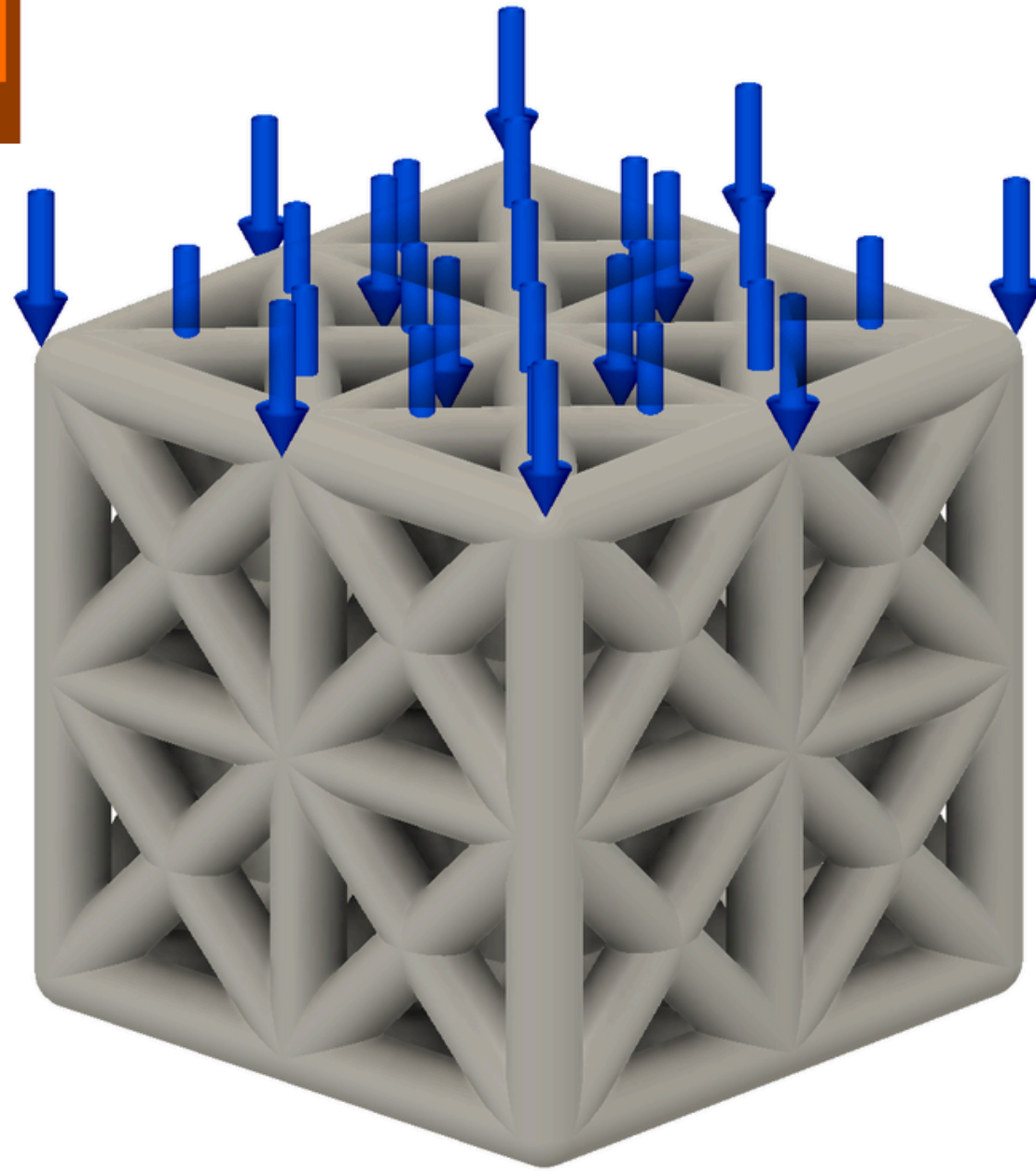
- Import into **DOLFINx**
- B.C (fixed bottom surface)
- Load applied on top surface
- Displacement, Von Mises stress and safety factor



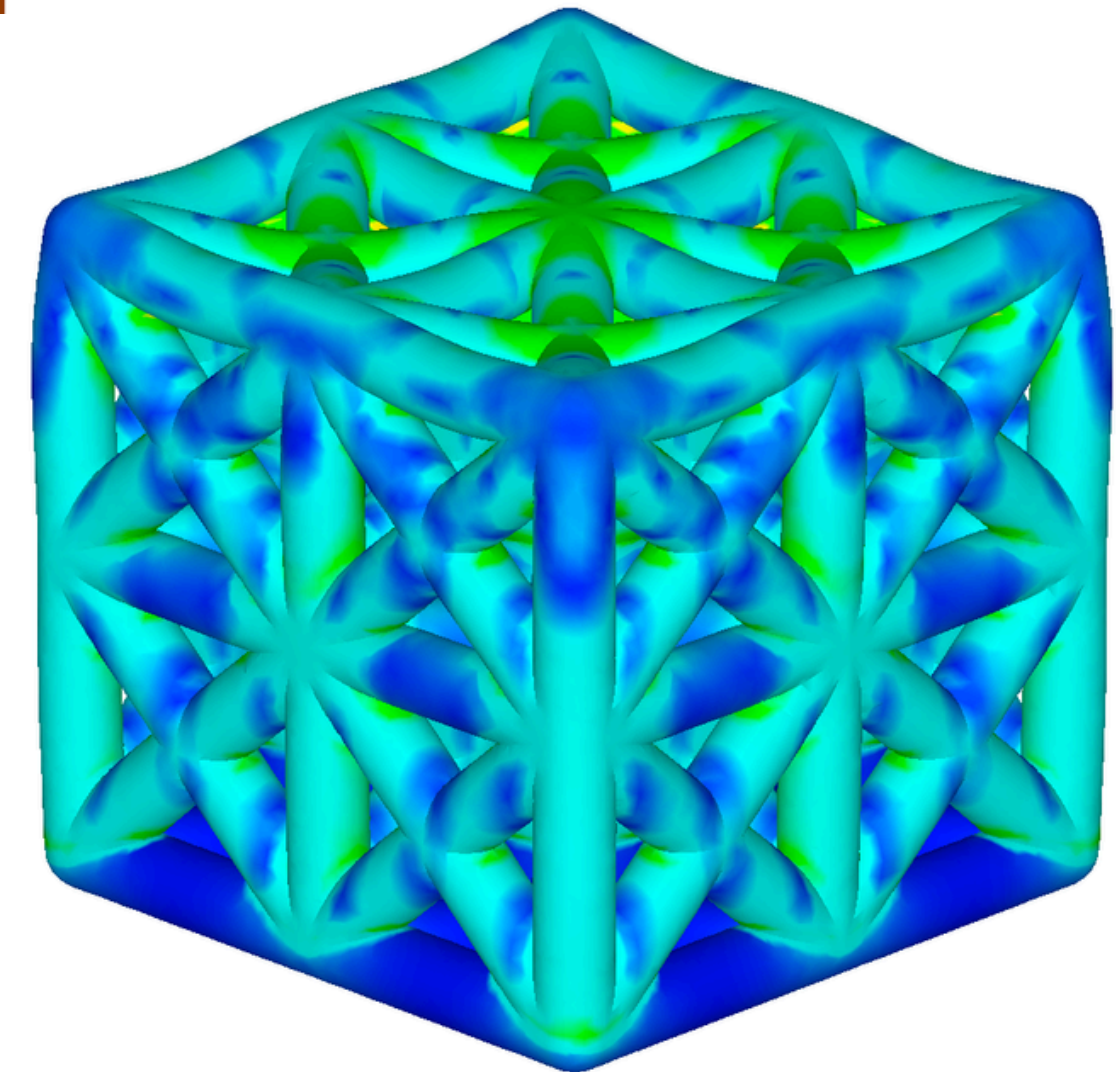
.pvd



Model Verification

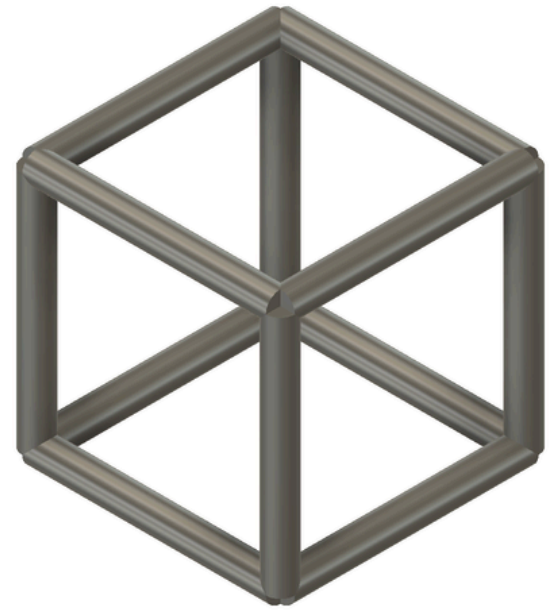


Simulation Model

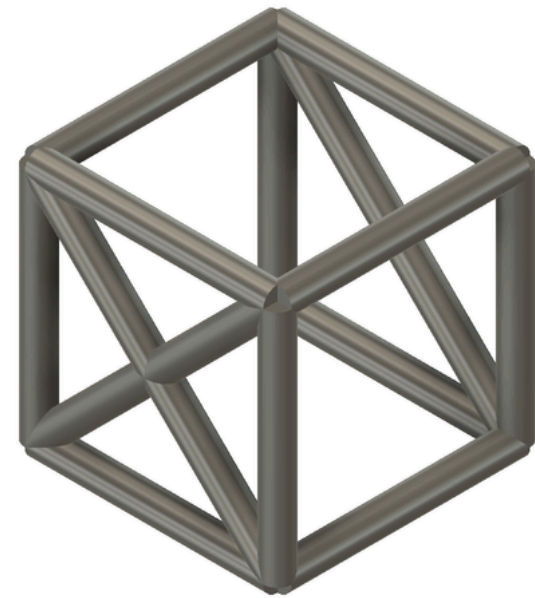


Simulation Result

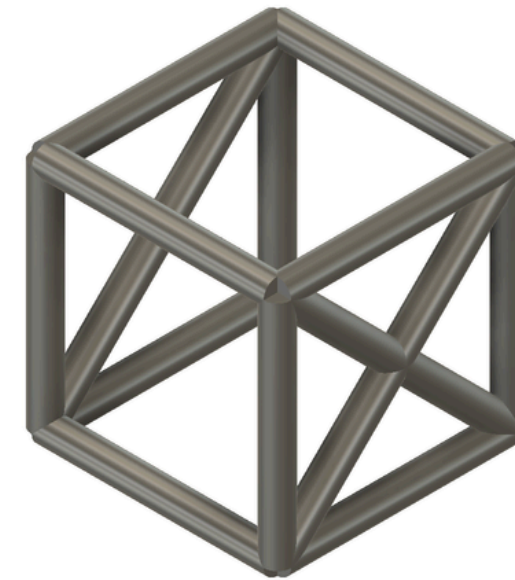
Data reduction



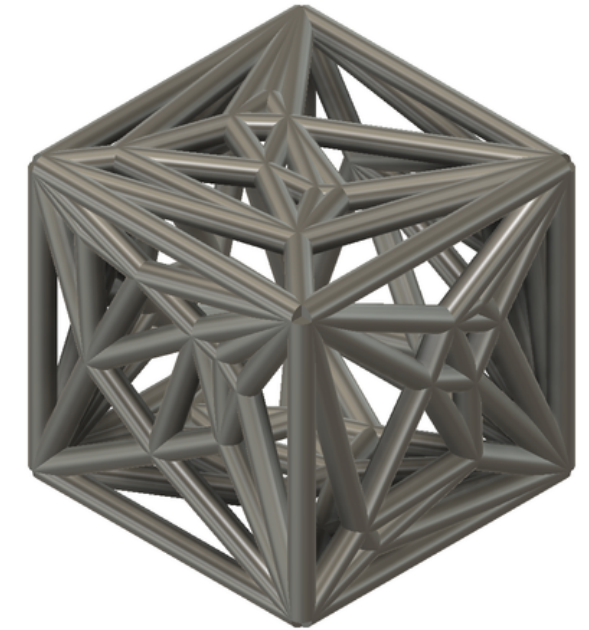
Baseline



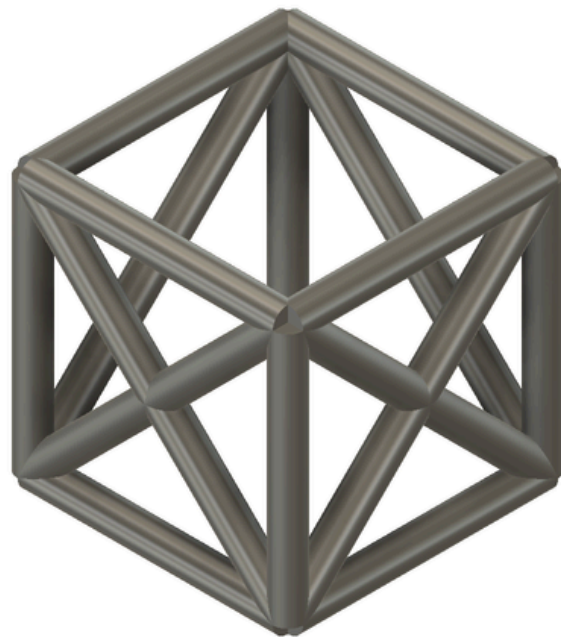
Lateral bracing (XZ)



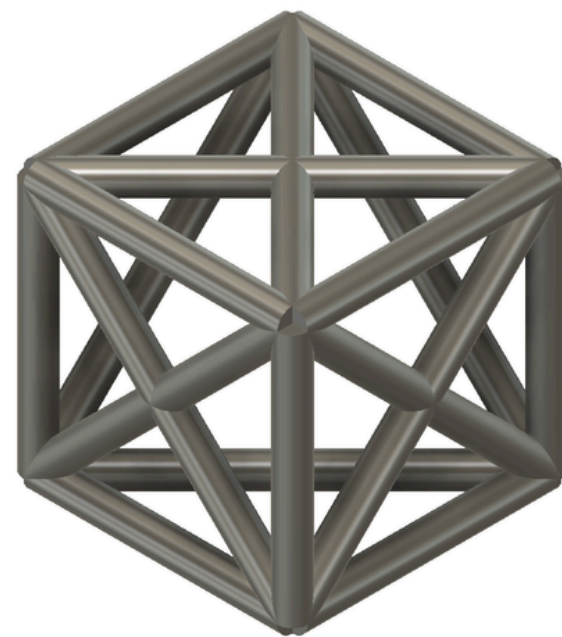
Lateral bracing (YZ)



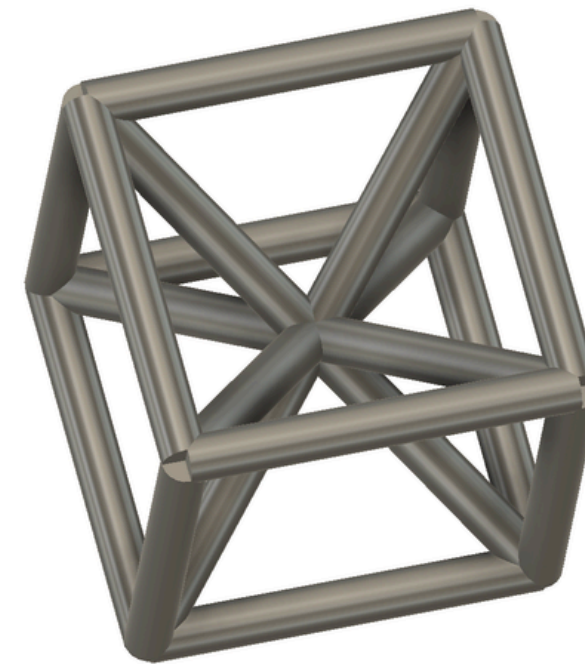
Random



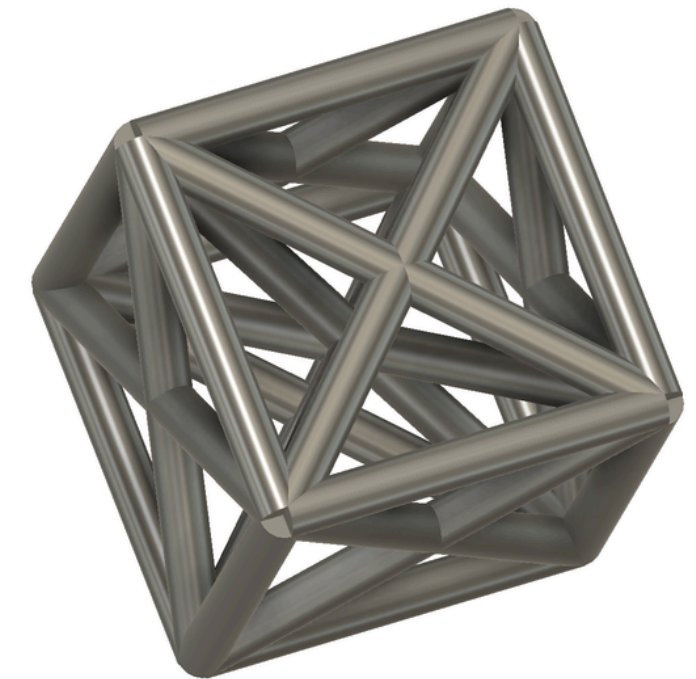
XZ + YZ



XY + XZ + YZ

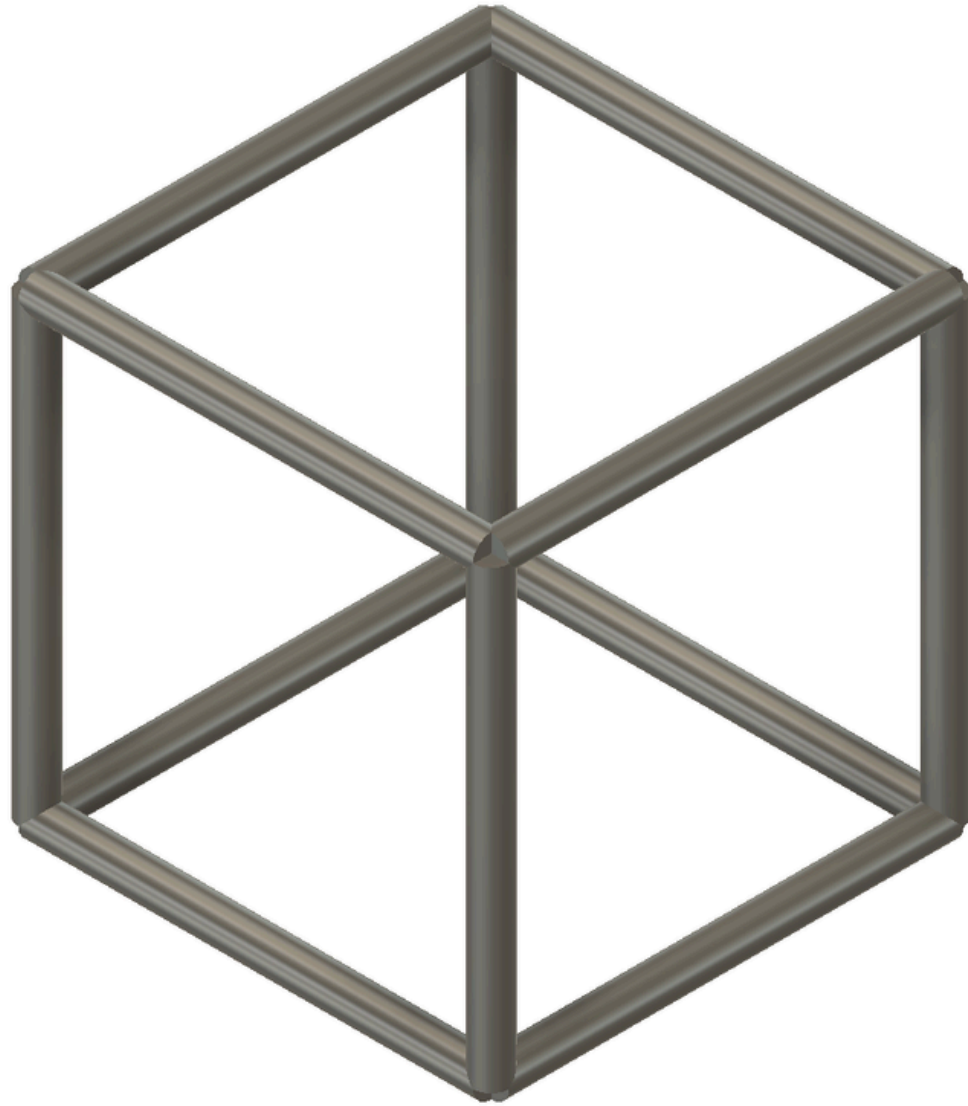


Body

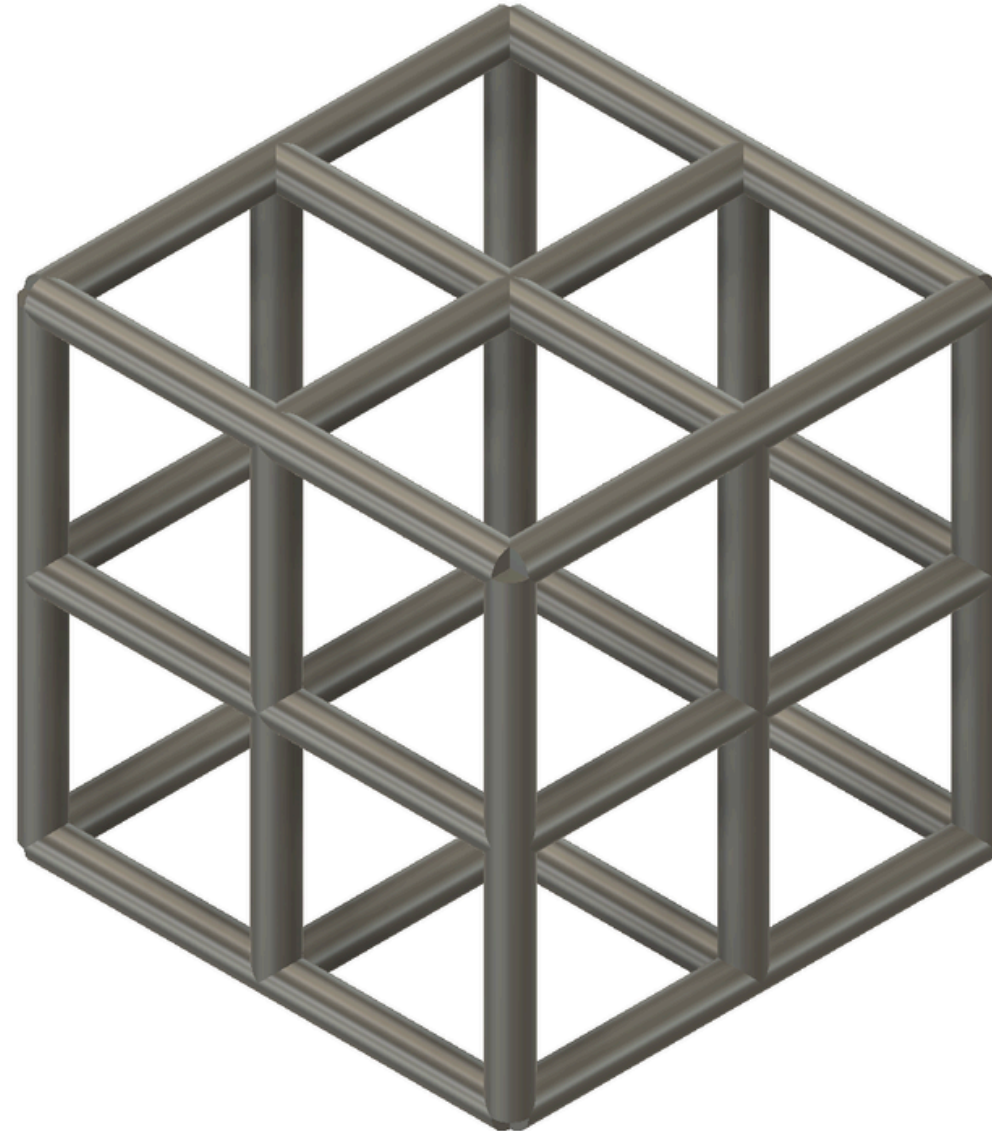


Full

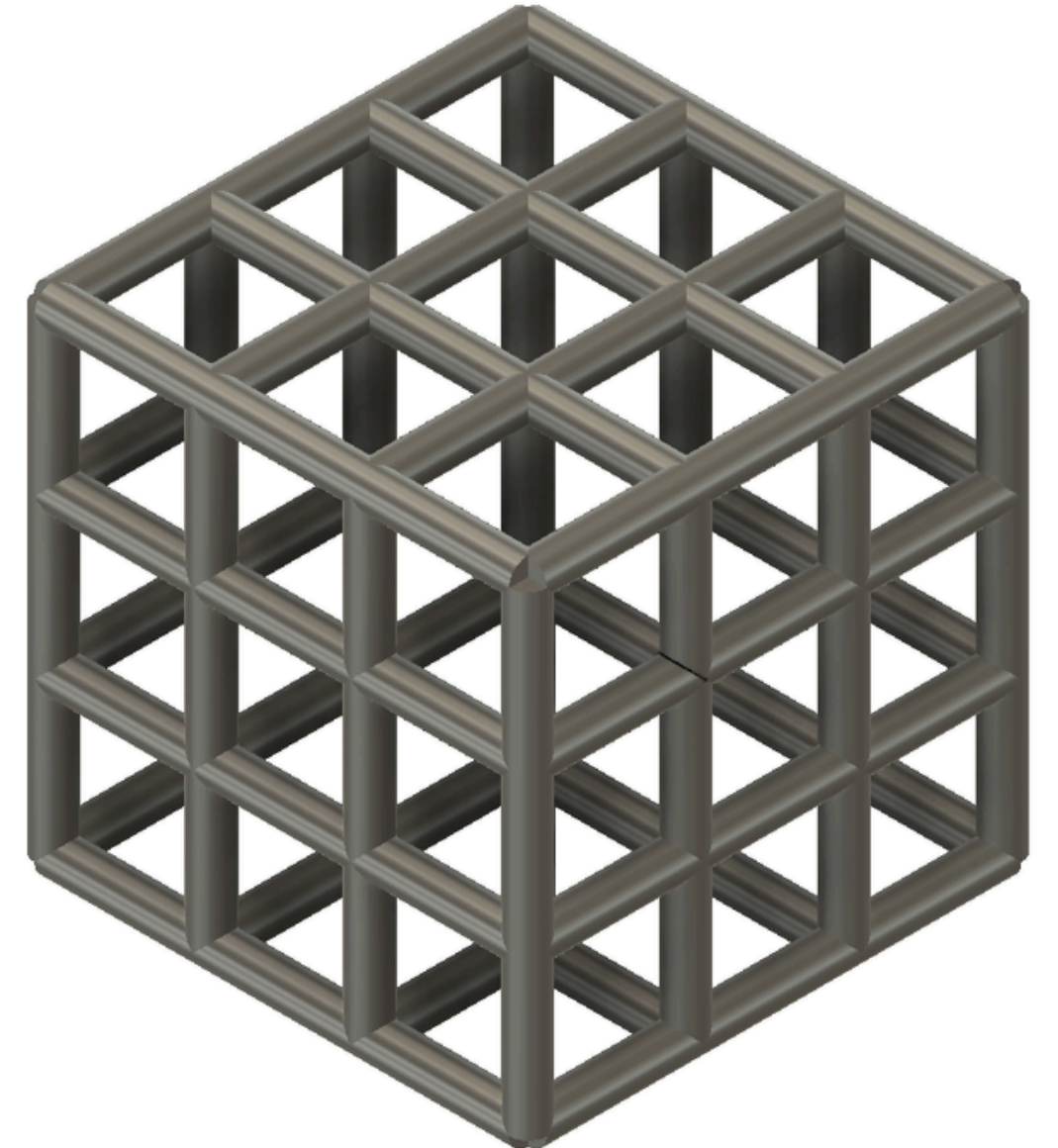
Data reduction



$N = 1$

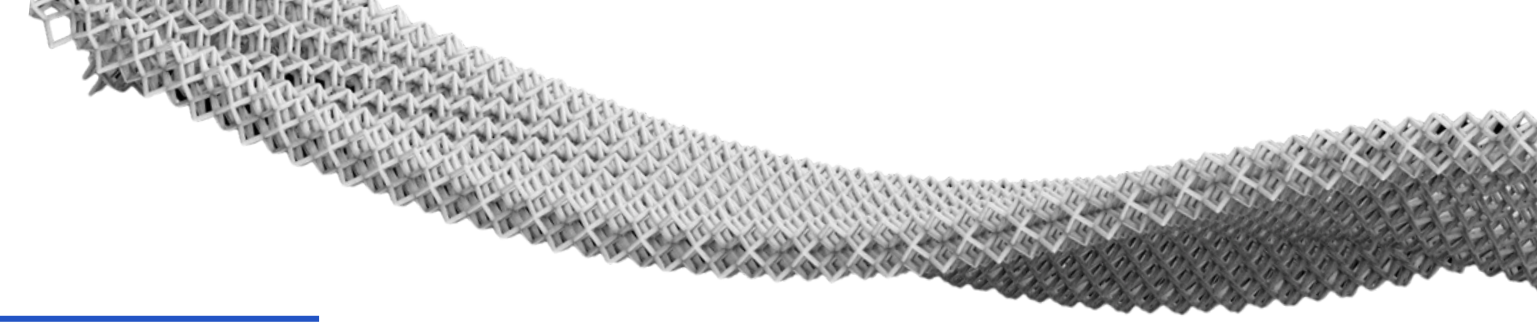


$N = 2$



$N = 3$

Design Plan



PROBLEM FORMULATION

$$\min_{\mathbf{x}} \mathbf{F}(\mathbf{x}) = [m(\mathbf{x}), \sigma_{VM}^{\max}(\mathbf{x})]$$

$$\mathbf{x} = (r, n, t) \in \mathbb{R} \times \mathbb{Z} \times T$$

$$\text{s.t.} \quad r \leq 0.45 \cdot \frac{a}{n} \quad a = 50\text{mm}$$

DESIGN SPACE

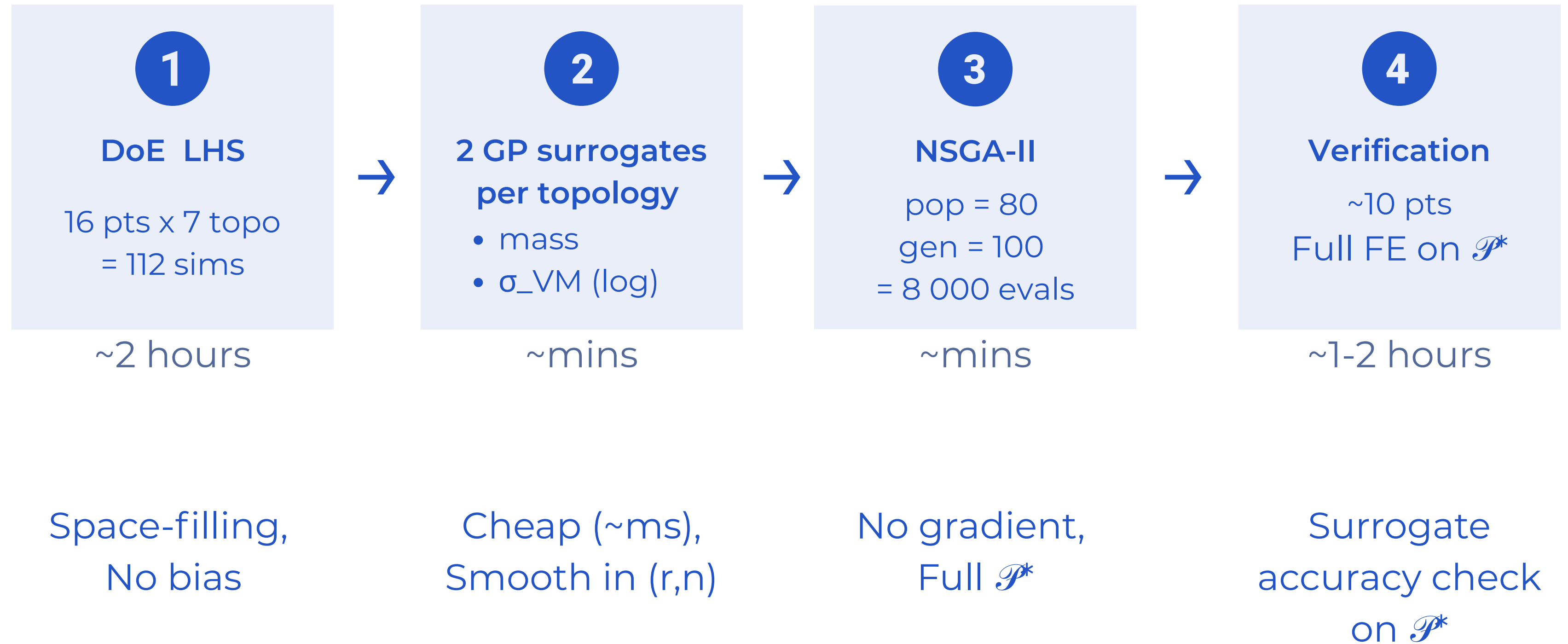
- $r \in [1.0, 2.0]\text{mm}$ ($\mathbb{R}$)
- $n \in \{1, 2, 3\}$ ($\mathbb{Z}$)
- $t \in T, |T| = 7$ (categ.)

CHALLENGES

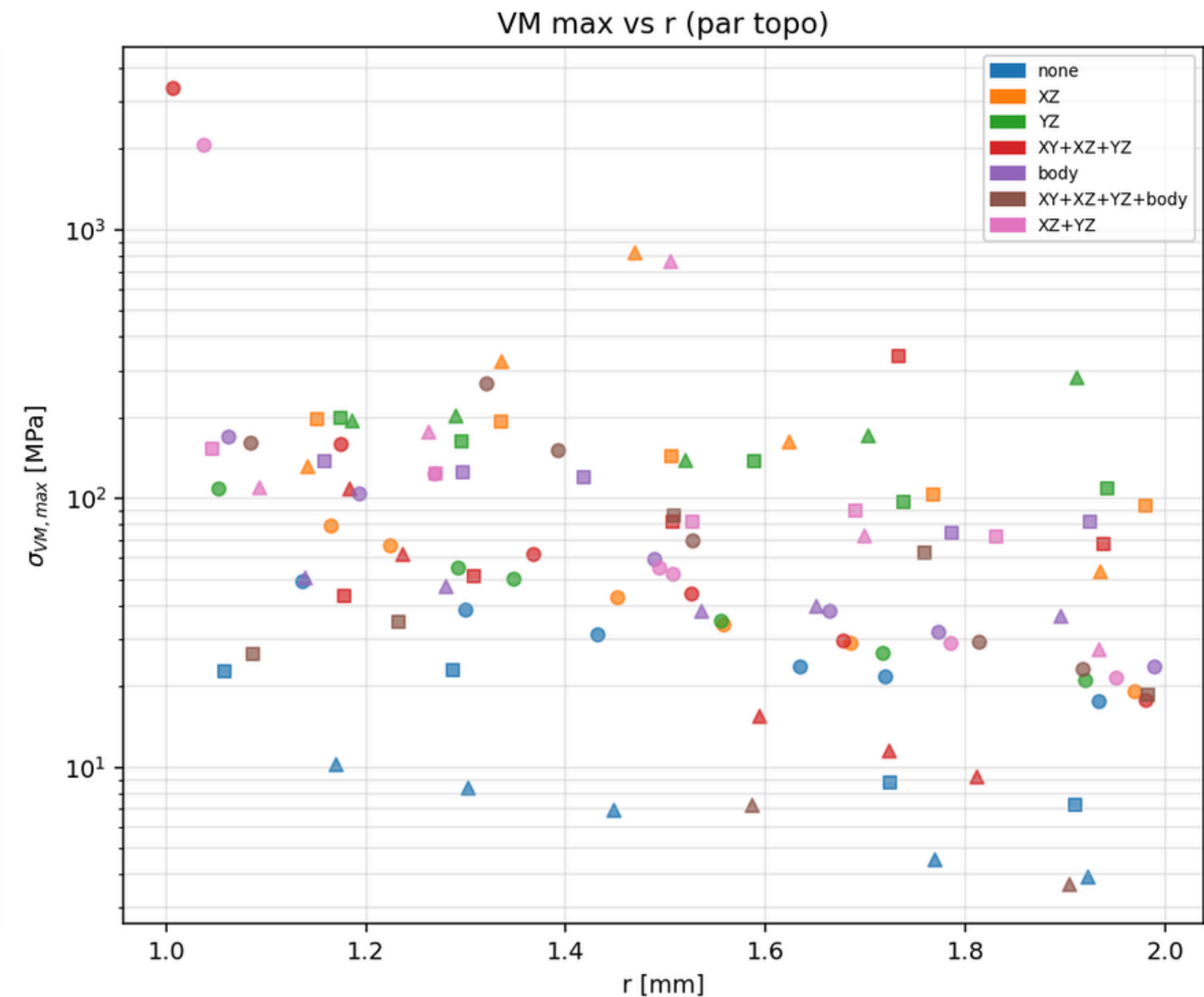
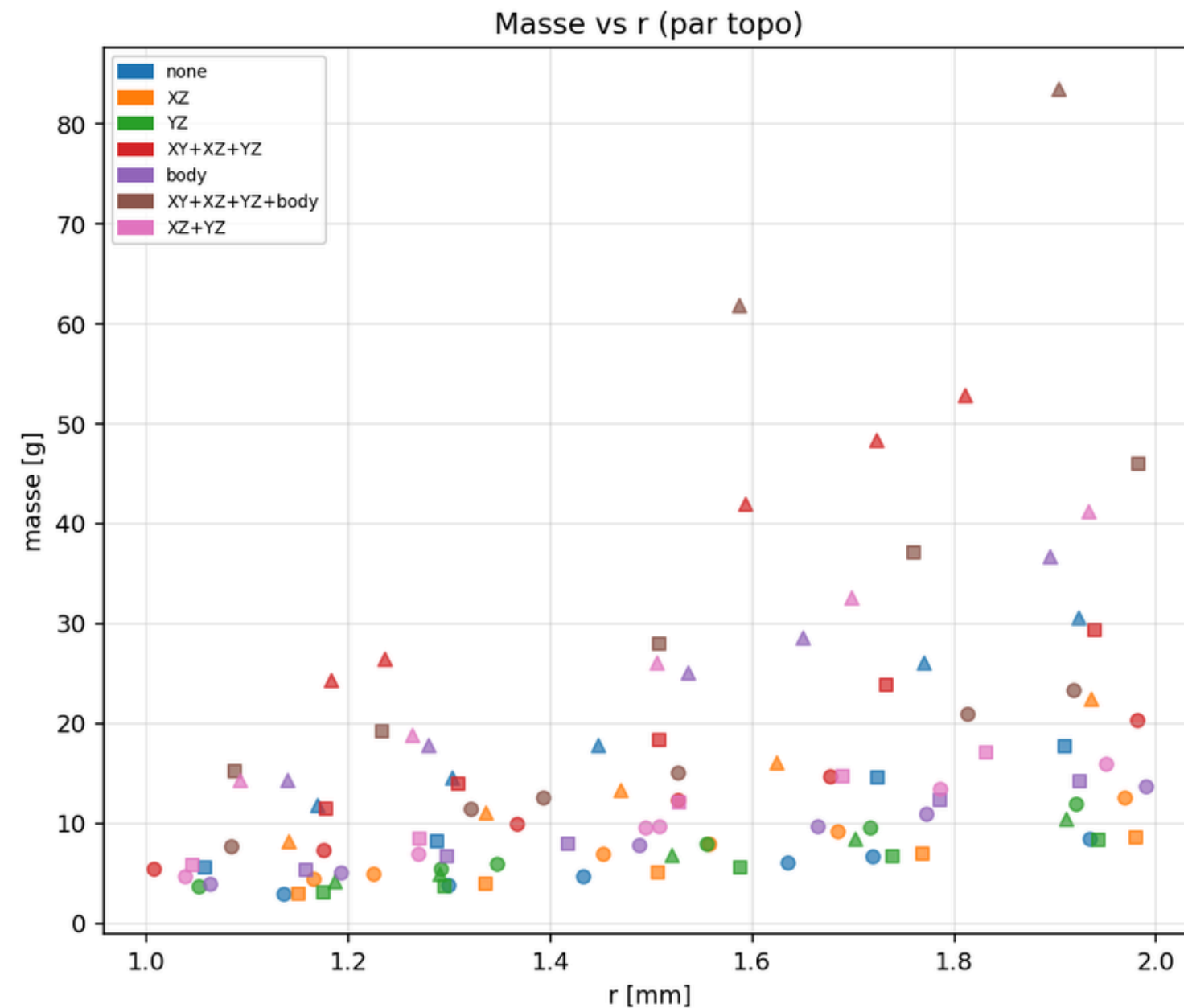
- **Mixed $\mathbb{R} \times \mathbb{Z} \times \text{categ. space}$** + non-convex
 - **Expensive black-box:** $\sim \text{min/eval}$, ∇F unknown
 - **Multi-objective:** Full $\mathcal{P}^*$ needed
- Gradient methods unsuitable
→ Surrogate needed
→ Pareto-aware algorithm

⇒ **Design Space → DoE → Surrogate → Search (Pareto front) → Solution**

Optimization Plan



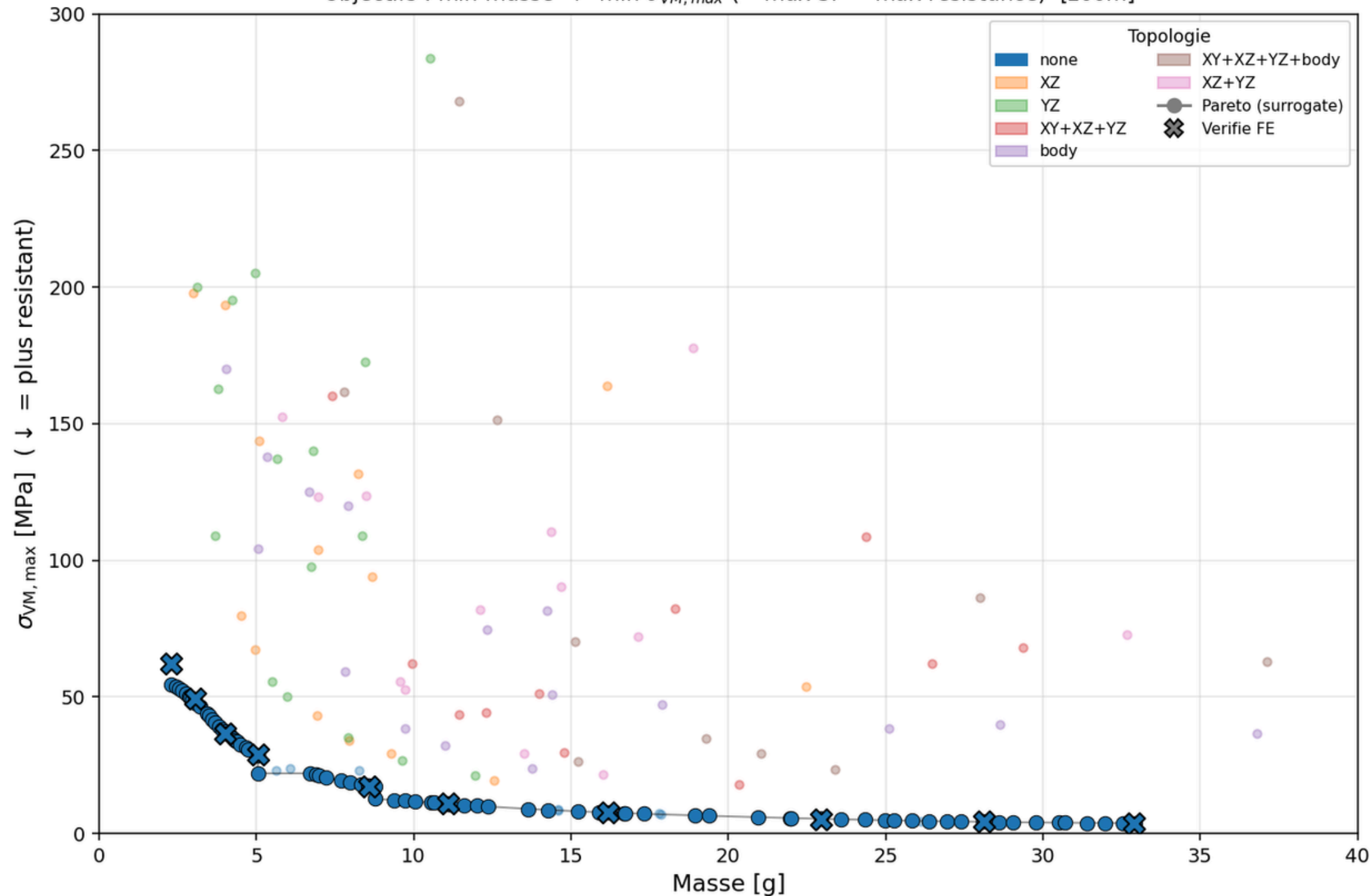
Optimization results



- **Mass** grows **monotonically** with **r** and **topological richness**
- **$\sigma_{VM,max}$** , **no trend with r** → non-trivial response surface
- Low-r / rich-topology **outliers** → motivate P97 filter

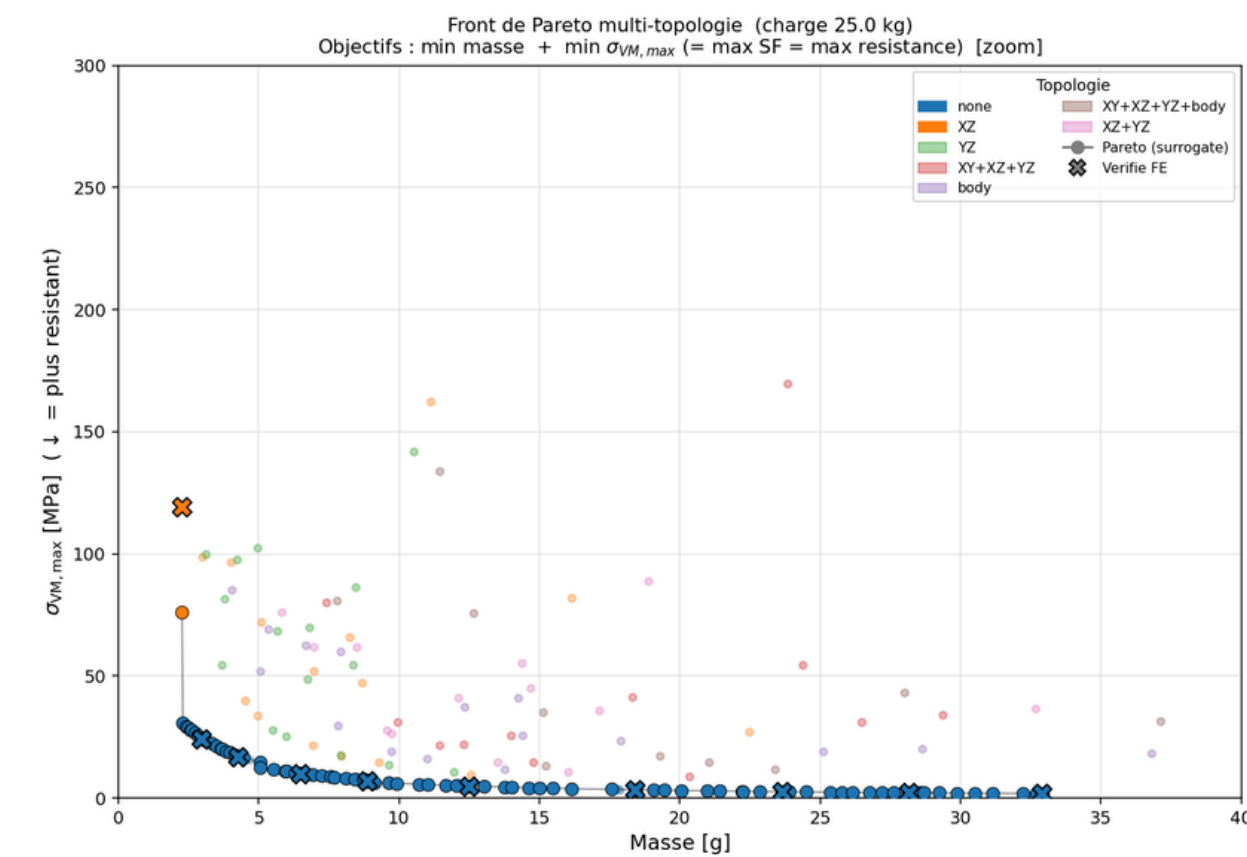
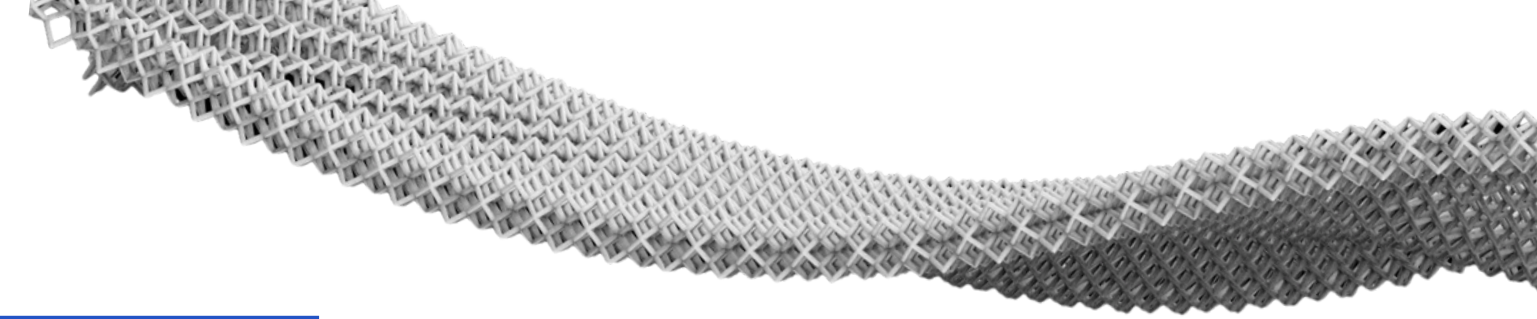
Optimization results

Front de Pareto multi-topologie (charge 50.0 kg)
Objectifs : min masse + min $\sigma_{VM,max}$ (= max SF = max resistance) [zoom]

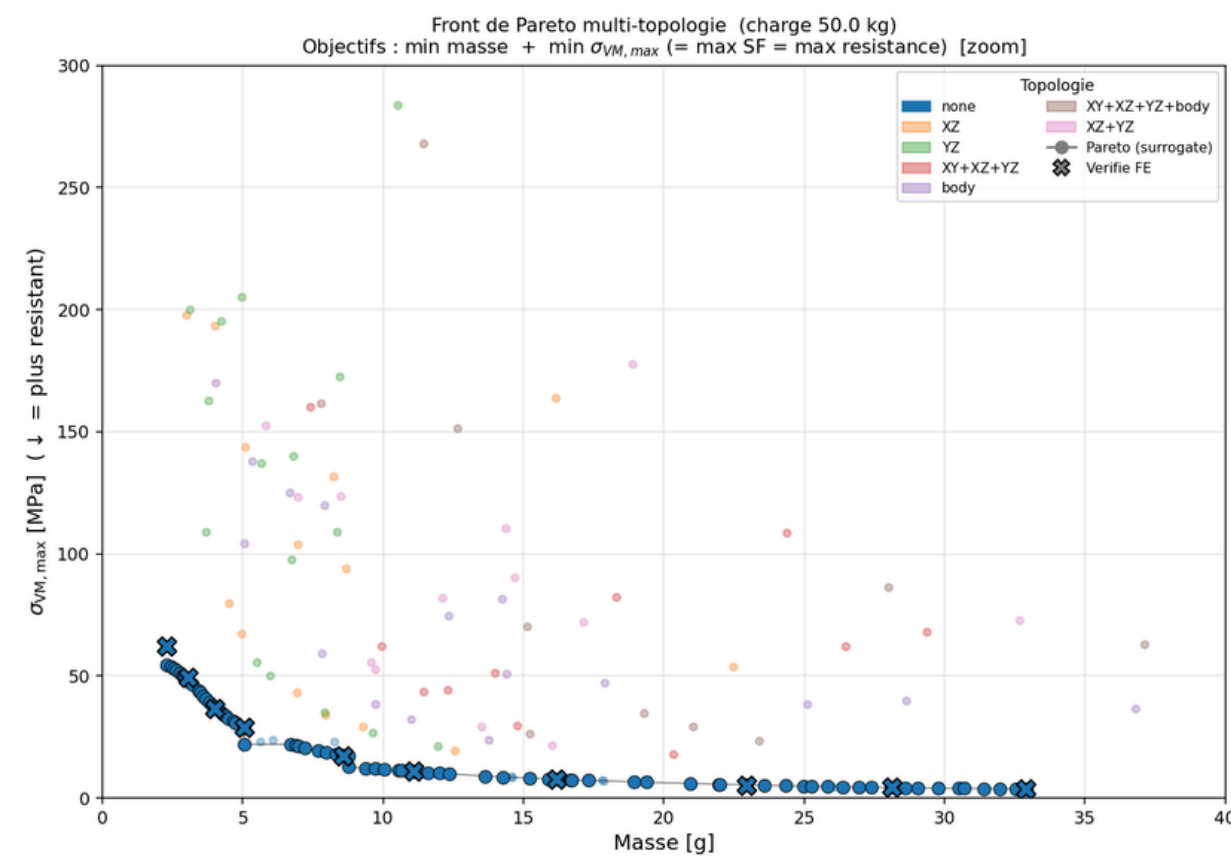


- **None** topology dominates the entire useful front (2–33 g)
- **Rich topologies** appear only at **high mass** with **no stress benefit**
- **Extreme outliers** (>2000 MPa at low mass) → numerical collapse, motivates P97 filter
- FE verification (X) tracks surrogate front closely

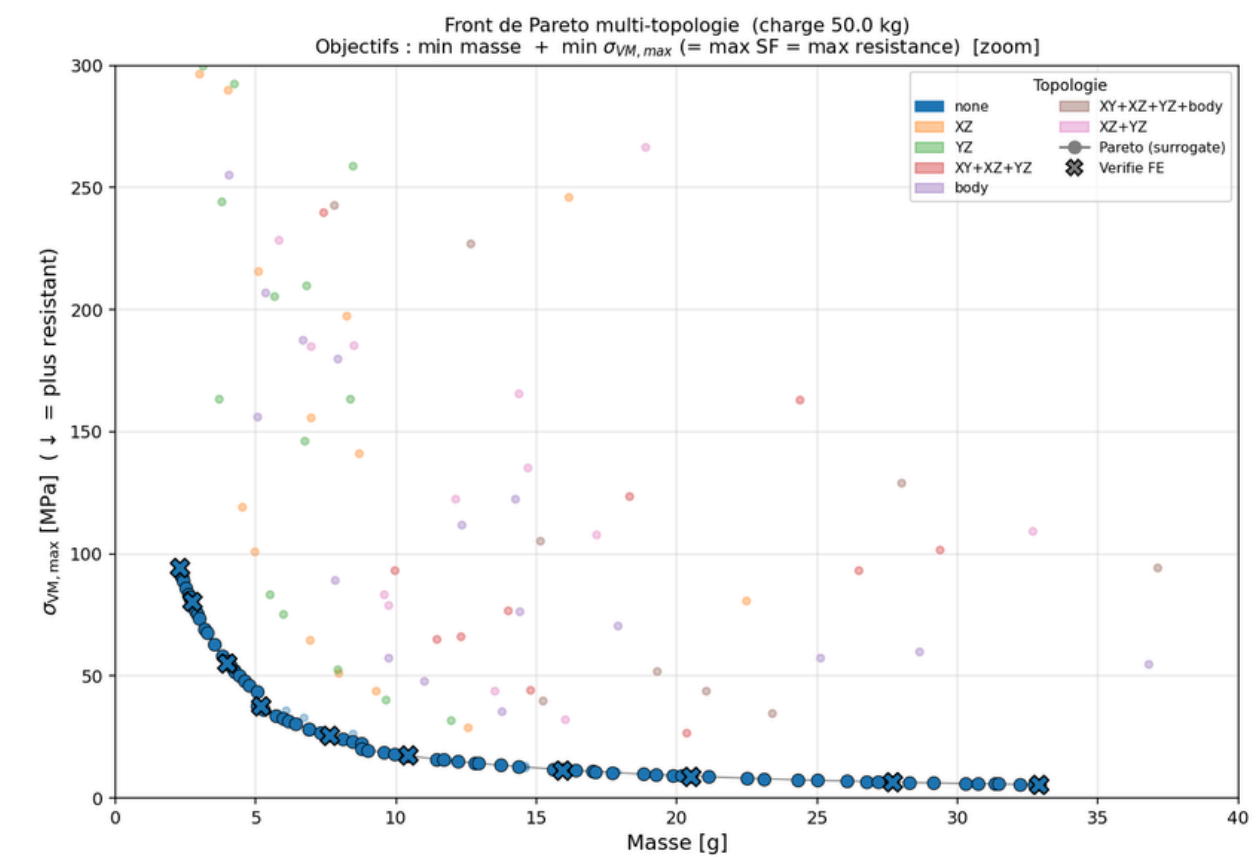
Load Cases



25 kg



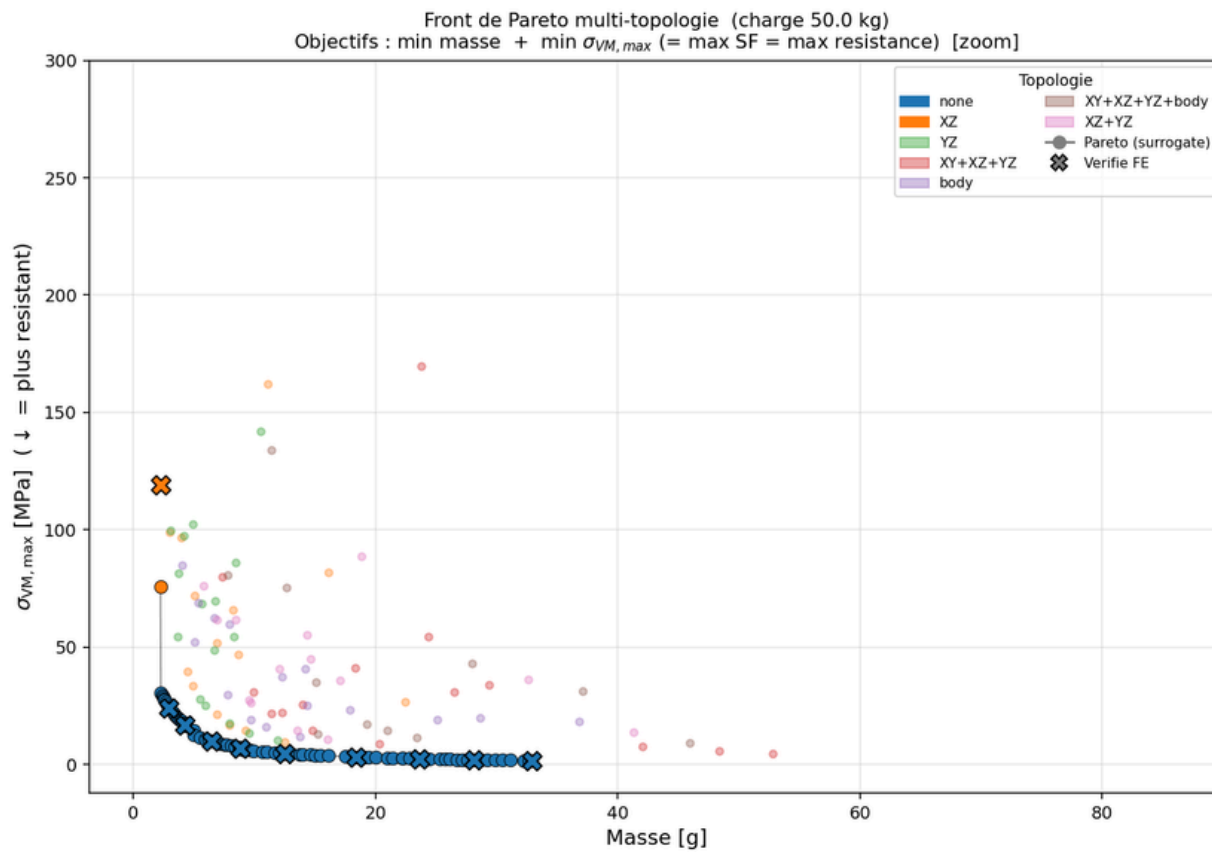
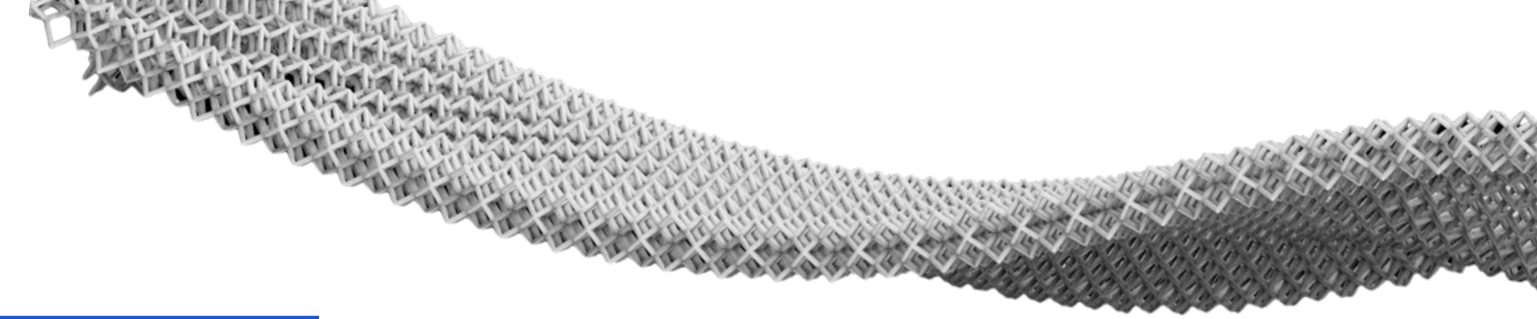
50 kg



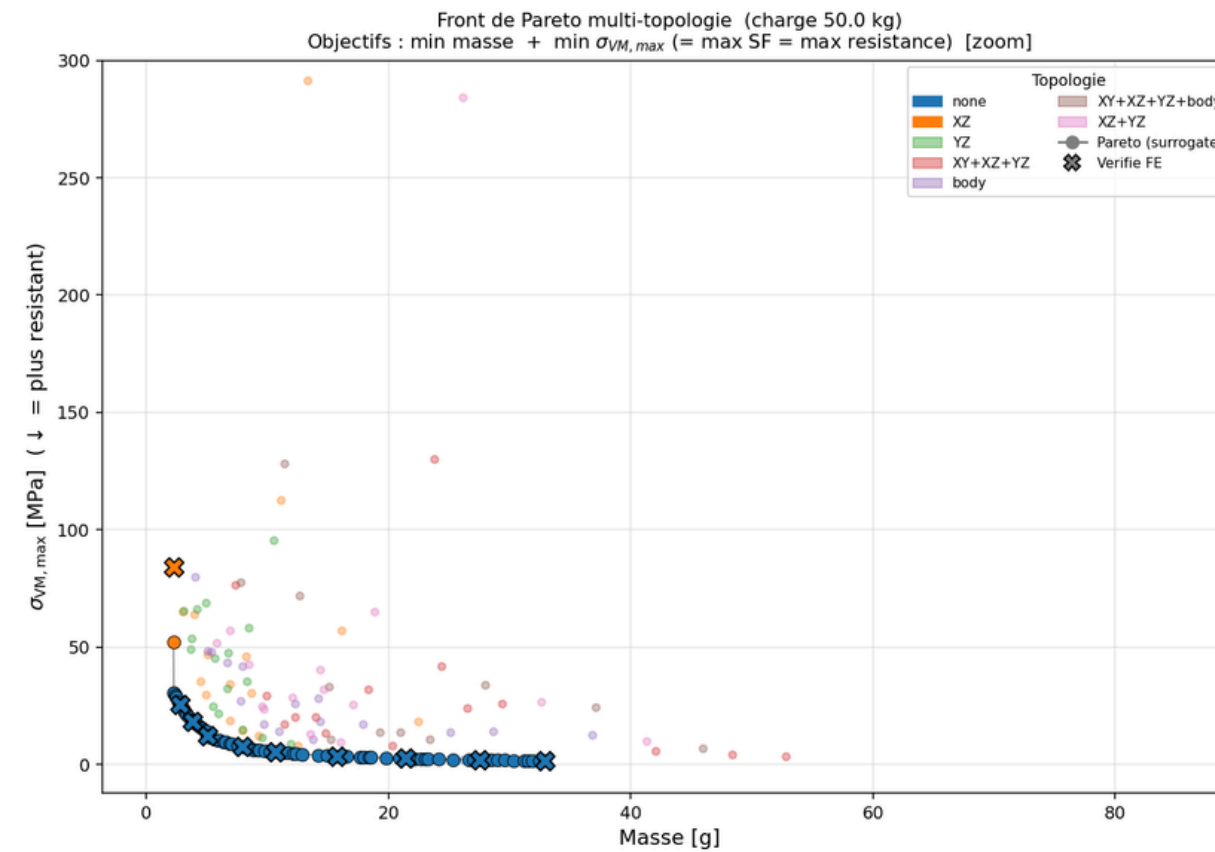
75 kg

- σ_{VM} scales ~linearly with load
- Front shape and none dominance preserved across all three loads
- Useful design region stays at 2–33 g; only outliers shift upward
→ optimal topology family is robust to load magnitude

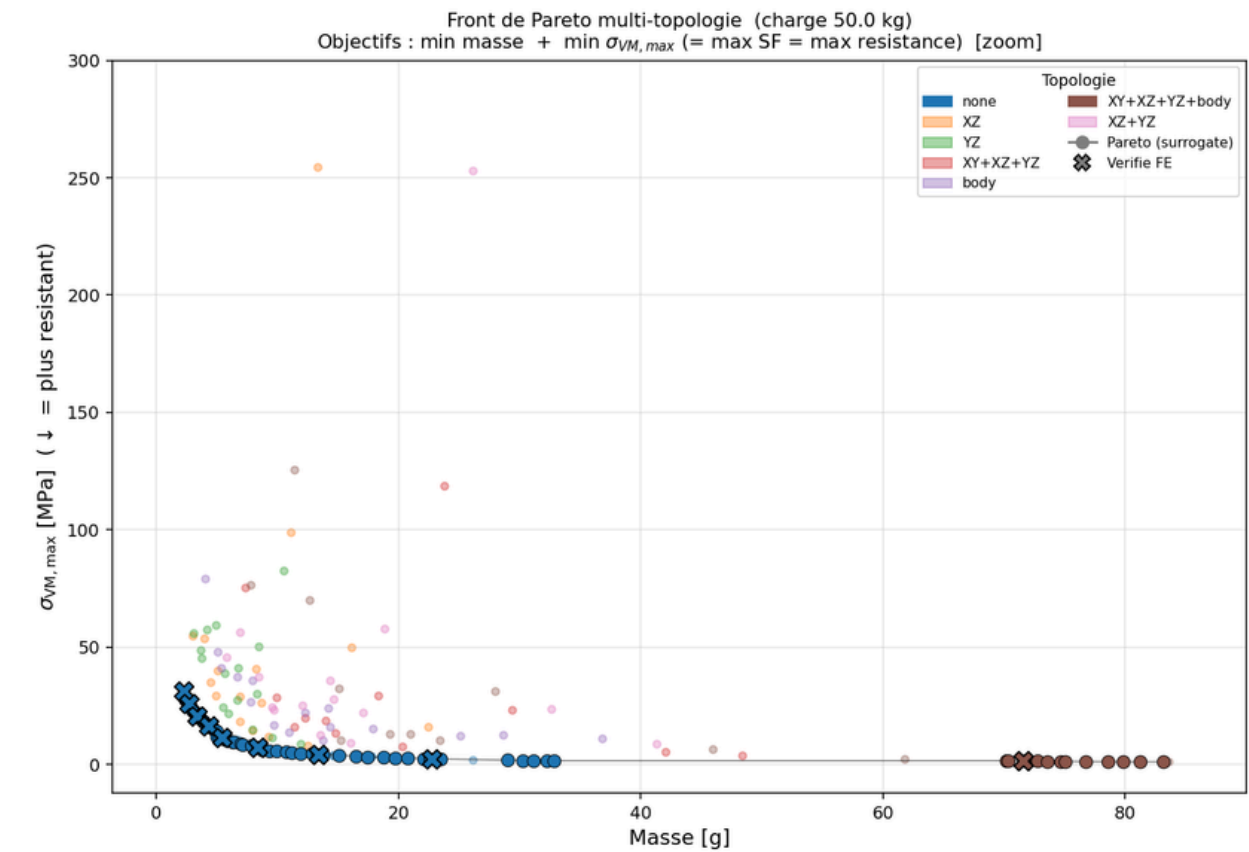
Stress Criterion



σ_{max}



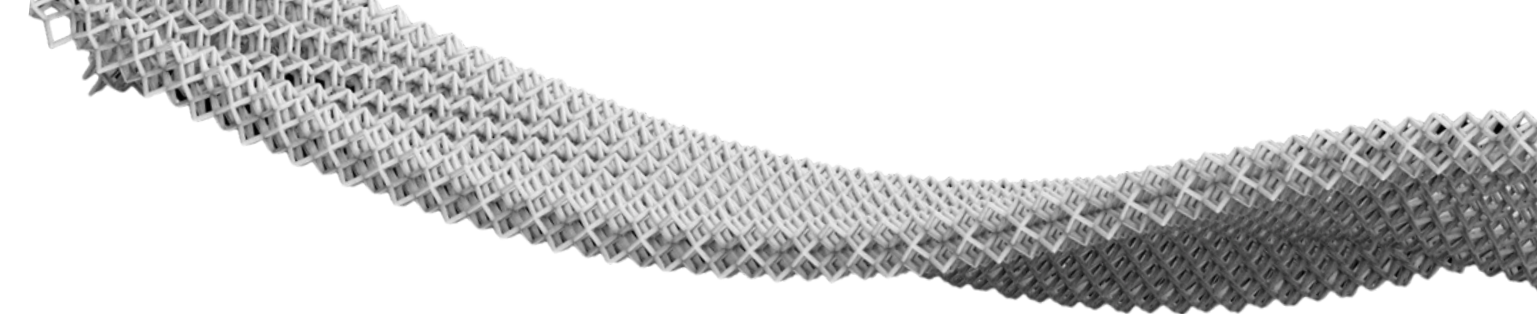
σ_{P97}



σ_{P95}

- Presence of stress concentration $\rightarrow$ need of this stress criterion to remove the concentrators outliers at 25 kg
- Front shape and none dominance unchanged between max, P97 and P95
- P95 highlight the high performance of other heavy topology

Final result



score = 0,75 · masse_norm + 0,25 · σ_vm_norm

→ The smallest, the better the design

For 25 kg, with σ_P95

#	r (mm)	n	masse (g)	σ_vm (MPa)	Score
1	1.264	2	7.98	7.84	0.199
2	1.107	3	10.68	5.47	0.233
3	1.301	1	3.87	18.29	0.25
4	1.355	3	15.73	3.65	0.336
5	1.593	3	21.41	2.67	0.467
6	1.819	3	27.56	2.09	0.618

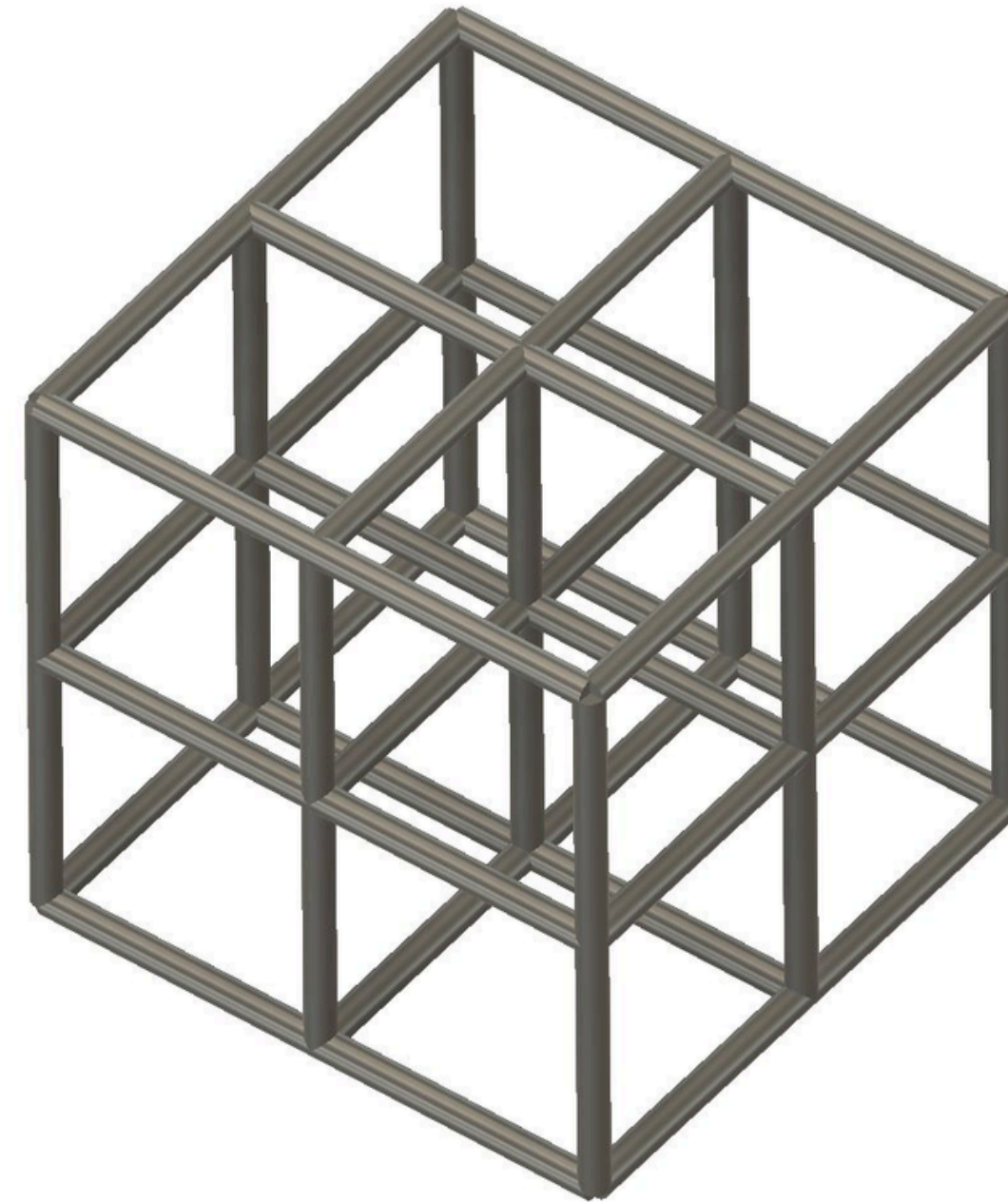
Final result

PARAMETERS

- $r = 1.264$
- $n = 2$
- Topology = none

RESULTS

- mass = 7.98g
- $\sigma_{VM} \max = 7.84 \text{ MPa}$
- SF = 3.82



Thank You

